NEXORA SYSTEMS

Fictional Company Knowledge Base

RAG Testing & Practice Corpus

Purpose

This document is intentionally fictional. It is designed to be ingested into a Retrieval-Augmented Generation system so you can practice document extraction, chunking, embeddings, retrieval, metadata filtering, citations, and question answering.

Do not treat any information in this document as real-world company information.

1. Company Overview

Nexora Systems is a fictional cloud software company founded in 2019 by Maya Chen and Daniel Ribeiro. Its headquarters are in Lisbon, Portugal, with engineering offices in Porto, Dublin, and São Paulo.

Nexora develops software for business process automation. Its main product, **NexoraFlow**, allows organizations to automate approvals, notifications, data synchronization, scheduled jobs, and event-driven workflows.

As of January 2026, Nexora has approximately 286 employees. The company is divided into Engineering, Product, Sales, Customer Success, Support, Finance, Security, Legal, and People Operations.

The company's fictional mission is: *“Make complex business operations programmable for every team.”*

Nexora's stated values are reliability, simplicity, transparency, customer empathy, experimentation, and security.

The main corporate website used in this training corpus is **nexora.example**. All email addresses in this document use the fictional **nexora.example** domain.

2. Leadership and Departments

The Chief Executive Officer is Maya Chen. The Chief Technology Officer is Daniel Ribeiro. The Chief Financial Officer is Sofia Almeida. The Chief Security Officer is Omar Haddad. The Vice President of Customer Success is Laura Bennett.

Engineering is led by Daniel Ribeiro and is divided into Platform Engineering, Application Engineering, Infrastructure, Developer Experience, and Data Systems.

The Security team owns security monitoring, access reviews, vulnerability management, incident response, and security training.

Customer Success manages onboarding, adoption, renewals, and enterprise customer relationships.

The Support organization is responsible for technical support, ticket triage, troubleshooting, and escalation.

3. Products

NexoraFlow is available in three plans: Launch, Growth, and Enterprise.

Launch is intended for small teams and includes 8 active workflows, 5 users, scheduled workflows, standard connectors, and community support.

Growth includes 60 active workflows, 30 users, advanced connectors, workflow version history, audit logs, team permissions, and priority support.

Enterprise includes unlimited workflows subject to fair-use policies, SSO, SCIM, advanced RBAC, custom data retention, private networking options, dedicated support, and contractual service-level commitments.

NexoraForms is an internal-form product for creating approval forms and data-entry interfaces. NexoraPulse provides operational dashboards and workflow analytics.

The NexoraFlow API allows customers to create workflows, inspect workflow runs, manage connectors, and retrieve audit events.

4. Pricing

The following prices are fictional training data and are expressed in USD per workspace per month.

Plan	Monthly	Annual	Users	Active Workflows

Launch	$39	$390/year	5	8
Growth	$179	$1,790/year	30	60
Enterprise	Custom	Custom	Custom	Unlimited*

* Enterprise workflows are subject to fair-use and contractual limits.

Annual Launch billing is equivalent to ten monthly payments rather than twelve. Growth annual billing follows the same fictional convention.

Enterprise pricing is negotiated based on workflow volume, users, support requirements, security requirements, deployment architecture, and contractual terms.

The Growth plan includes a 21-day trial. No credit card is required to begin a trial. Trial workspaces become read-only 7 days after the trial ends unless the customer upgrades or exports its data.

5. Customer Support

Launch customers receive community support and standard documentation. Growth customers receive priority email support. Enterprise customers receive 24/7 critical-incident support and an assigned customer success manager.

Normal support hours are Monday through Friday from 08:00 to 18:00 Western European Time, excluding public holidays.

Support tickets should contain the organization ID, workspace ID, approximate incident time, workflow ID, error message, affected users, and relevant recent configuration changes.

Support severity levels are S1, S2, S3, and S4. S1 means a critical production outage affecting a major business function. S2 means substantial degradation. S3 means a normal defect or operational problem. S4 means a question, documentation request, or minor issue.

S1 incidents are escalated to the on-call engineering team. An initial customer update should normally be prepared within 30 minutes after confirmation.

6. Security

Nexora encrypts data in transit using TLS and encrypts stored customer data using managed cloud encryption.

Employees must use multi-factor authentication for corporate systems. Production access is role-based and elevated privileges require approval.

Customer API secrets are stored in a managed secrets service. Passwords are never stored as plaintext.

If a customer believes an API credential has been exposed, the recommended action is to revoke or rotate the credential immediately, review recent usage, and contact security@nexora.example.

The fictional company performs quarterly access reviews, continuous dependency scanning, annual penetration testing, backup restoration tests, and incident-response exercises.

Security incidents involving potential unauthorized access are escalated to the Security team and handled according to contractual obligations and applicable law.

7. Privacy and Data Retention

Nexora processes customer data to provide services, execute workflows, maintain security, provide support, and improve service reliability.

Standard workflow execution logs are retained for 120 days. Enterprise customers may request retention periods of 30, 120, 240, or 365 days.

Deleted data may remain temporarily in encrypted backups until the backup lifecycle expires.

Customers are responsible for determining whether their data is appropriate for the service and for configuring user permissions.

Privacy questions can be sent to privacy@nexora.example. Security questions should be sent to security@nexora.example.

8. Technical Architecture

NexoraFlow uses a distributed cloud architecture. Incoming API requests pass through an authenticated API gateway and are routed to internal services.

Workflow execution is asynchronous. A trigger creates a job, the job enters a queue, and worker services execute workflow steps.

Transactional state is stored in a relational database. Large objects are stored in object storage. Logs and metrics are collected by the observability platform.

Development, staging, and production are isolated environments. Production deployments require automated tests, review, and an approved deployment process.

A workflow failure investigation normally begins with execution logs, workflow configuration, recent deployment changes, connector health, and service health.

9. API Documentation

The NexoraFlow API uses HTTPS and JSON. API authentication may use API keys or OAuth depending on the endpoint and integration.

HTTP 400 indicates a malformed request. HTTP 401 indicates authentication failure. HTTP 403 indicates insufficient permission. HTTP 404 indicates a missing resource. HTTP 409 indicates a conflicting state. HTTP 422 indicates validation failure. HTTP 429 indicates that a rate limit has been exceeded. HTTP 500 indicates an internal server error.

Clients receiving HTTP 429 should use exponential backoff and should avoid immediately retrying requests in a tight loop.

Webhook consumers should acknowledge events quickly and process long-running operations asynchronously. Webhook signatures must be validated before accepting events.

10. Engineering Practices

Production code changes require peer review. Engineers create pull requests, automated tests run in CI, and approved changes are merged into protected branches.

Emergency changes are permitted during active incidents but must be documented after the incident.

Every production incident must have an incident record containing impact, timeline, actions taken, mitigation, root-cause analysis, and follow-up tasks.

Engineering teams hold a weekly architecture review on Thursday afternoons. Platform Engineering owns the shared infrastructure layer, while Application Engineering owns customer-facing workflow features.

11. Sales and Procurement

Launch and Growth can be purchased online. Enterprise customers usually begin with a sales consultation.

The standard Enterprise sales process consists of discovery, technical evaluation, security review, legal review, commercial negotiation, and contract activation.

Enterprise procurement may require a security questionnaire, data-processing agreement, service-level agreement, purchase order, and legal review.

Sales representatives cannot approve non-standard contractual liability terms without Legal review.

12. Employee Policies

Nexora operates a hybrid work model. Individual teams publish their office schedules. Employees receive company-managed laptops.

Employees must not share corporate credentials. Suspected phishing should be reported to security@nexora.example.

Employees are encouraged to document important technical and operational decisions in the internal knowledge base.

The fictional annual learning budget is USD 1,200 per employee. Employees may use it for approved courses, conferences, books, and professional certifications.

13. Incident Response

The on-call engineer opens an incident record when a major production issue is confirmed. An incident commander coordinates the response.

The primary goal during an incident is to reduce customer impact. Possible mitigations include rollback, disabling a failing integration, traffic changes, workload reduction, or temporary feature restrictions.

After an incident is resolved, the team performs a post-incident review. The review focuses on system behavior, contributing factors, and corrective actions.

For an S1 incident, the first customer-facing update should normally be prepared within 30 minutes of confirmation. Subsequent updates should be provided as material information becomes available.

14. Offices and Time Zones

Headquarters: Lisbon, Portugal. Engineering office: Porto, Portugal. European support center: Dublin, Ireland. South American engineering office: São Paulo, Brazil.

The company primarily uses Western European Time for standard support operations. The São Paulo engineering team commonly works on Brasília Time.

Internal meetings that involve teams in more than two regions should be scheduled using the company calendar's automatic time-zone conversion.

15. Frequently Asked Questions

Q: How much does Launch cost?

A: $39 per month per workspace.

Q: How much does Growth cost?

A: $179 per month per workspace.

Q: How long is the Growth trial?

A: 21 days.

Q: Does the Growth trial require a credit card?

A: No.

Q: How long are standard workflow logs retained?

A: 120 days.

Q: What does HTTP 429 mean?

A: The API rate limit has been exceeded.

Q: Who should receive a privacy question?

A: privacy@nexora.example.

Q: Who should receive a security question?

A: security@nexora.example.

Q: What should be included in a support ticket?

A: Organization ID, workspace ID, approximate incident time, workflow ID, error message, affected users, and recent relevant configuration changes.

Q: What is an S1 incident?

A: A critical production outage affecting a major business function.

Q: How quickly should the first S1 customer update be prepared?

A: Normally within 30 minutes after confirmation.

Q: How much is the annual learning budget?

A: USD 1,200 per employee.

Q: Where is the headquarters?

A: Lisbon, Portugal.

Q: Which product provides operational dashboards?

A: NexoraPulse.

Q: Which product provides internal forms?

A: NexoraForms.

16. RAG Challenge Data

The following information is intentionally distributed across the document so that retrieval can be tested. Some questions require finding a single section; others require combining facts from different sections.

Challenge A: A customer wants to know the Growth price, trial length, and annual billing behavior. These facts are located in the Pricing section.

Challenge B: A customer reports HTTP 429 responses while a workflow is running. The API section explains the meaning of 429 and recommends exponential backoff. The Technical Architecture section explains that workflow execution is asynchronous.

Challenge C: A customer reports a possible leaked API credential. The Security section gives the immediate response, while the Privacy section distinguishes security contact information from privacy contact information.

Challenge D: An Enterprise customer asks about critical support. The Support section describes 24/7 critical-incident support and the S1 response process.

Challenge E: A prospective employee asks about remote work and professional development. The Employee Policies section contains the hybrid-work policy and annual learning budget.

Challenge F: A technical question asks what happens after an event triggers a workflow. The Technical Architecture section describes the trigger, queue, and worker-service sequence.

Challenge G: A procurement question asks which Enterprise documents may be required. The Sales and Procurement section lists security questionnaires, data-processing agreements, service-level agreements, purchase orders, and legal review.

17. Suggested RAG Test Questions

Use these questions to test retrieval quality. For each question, inspect the retrieved chunks before looking at the generated answer.

1. What is the monthly price of the Growth plan?
2. How long is the Growth trial?
3. Does the trial require a credit card?
4. How much does Launch cost?
5. What is the standard workflow log retention period?
6. What should a support ticket contain?
7. What does HTTP 429 mean?
8. How should a client respond to HTTP 429?
9. What does HTTP 422 mean?
10. What should a customer do if an API credential is exposed?
11. What is the security team's email address?
12. What is the privacy team's email address?
13. What is an S1 incident?
14. How quickly should the first S1 update be prepared?
15. Who is the CTO?
16. Where is Nexora's headquarters?
17. How many employees does Nexora have?
18. What product provides dashboards?
19. What product provides forms?
20. How does asynchronous workflow execution work?
21. What are the stages of Enterprise procurement?
22. What documents may be required during Enterprise procurement?
23. What is the employee learning budget?
24. What is the company's work model?
25. What should employees do about phishing?
26. What happens after a production incident?
27. What is the difference between 401 and 403?
28. What is the difference between 409 and 422?
29. What authentication methods does the API support?
30. What happens when a Growth trial ends?

18. Questions That Should Produce an 'Unknown' Answer

These are deliberately absent from the company knowledge base. A properly grounded RAG system should not invent answers.

1. What is the salary of a senior software engineer?
2. Who is the head of the Quantum Computing division?
3. What is Nexora's stock ticker?
4. How many patents does Nexora own?
5. What is the CEO's home address?
6. What is the exact database vendor used by Nexora?
7. What is the office rent in Lisbon?
8. What was Nexora's revenue in 2025?
9. What is the CEO's personal phone number?
10. What is the company's employee turnover rate?

19. Recommended RAG Experiments

Experiment 1 — Chunk Size: compare 400, 800, 1200, and 1600 characters.

Experiment 2 — Chunk Overlap: compare 0, 100, 200, and 300 characters.

Experiment 3 — Top-k: compare k=2, 3, 5, and 8.

Experiment 4 — Metadata: add department and section fields and filter retrieval by department.

Experiment 5 — Citations: make the model report the section used for each answer.

Experiment 6 — Unknown detection: introduce a similarity threshold and test the unknown questions.

Experiment 7 — PDF extraction: use PyMuPDF to extract this document and preserve page numbers in metadata.

Experiment 8 — Hybrid retrieval: combine keyword matching with embedding similarity.

Experiment 9 — Reranking: retrieve 15 candidates and rerank them down to 5.

Experiment 10 — FastAPI: expose the RAG system through a /ask endpoint.

Experiment 11 — Conversation: ask follow-up questions such as 'What about annual billing?' and add conversation-aware query rewriting.

Experiment 12 — Evaluation: create an answer key and calculate Recall@3 for the 30 test questions.